A Dirichlet polynomial with exact maximum N

Conjecture 00000001916

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We interpret the conjectured explicit factor as a constant independent of N, as required for the stated growth scale. An unrestricted N-dependent factor would impose no bound.

Theorem 1. *For the admissible coefficients $a_n = 1$,*

$$D_N(t) = \sum_{n=1}^{N} n^{it}, \qquad \sup_{t\in\mathbb{R}} |D_N(t)| = N.$$

This maximum is not $O\big(N^{1/2}(\log\log N)^{1/4}\big)$ as $N \to \infty$.

Proof. Every coefficient has modulus one. For real t, $n^{it} = \exp(it\log n)$ has modulus one, so the triangle inequality gives $|D_N(t)| \le N$. At $t = 0$ each term is one, and equality holds. Thus the supremum is attained and is exactly N.

For sufficiently large real x, $\log x \ge 1$ and

$$0 \le (\log\log x)^{1/4} \le (\log x)^{1/4} = o(x^{1/2}).$$

Multiplication by $x^{1/2}$ shows that the claimed scale is $o(x)$. Restricting to positive integers preserves this relation. If N were bounded by a constant times the claimed scale, transitivity would give $N = o(N)$, which is impossible: its defining estimate with constant $1/2$ would give $N \le N/2$ for all sufficiently large positive integers. □

Lean correspondence. `dirichlet` is the actual finite complex exponential sum, indexed by $n+1$ to avoid the logarithm of zero. `maximum_exact` proves the `IsGreatest` property for the range of its norm, so no informal supremum computation is assumed. `scale_littleO` uses the proved logarithm–power asymptotics of Mathlib, and `no_claimed_bound` negates the exact big-O assertion. The witness is deterministic, as in the question; no random-coefficient hypothesis appears in its text.

Verification. Lean 4.19.0 with Mathlib pinned to `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The accompanying project and verification transcript give the exact theorem and axiom audit. No proof placeholders, native evaluator proof bypass, or custom axiom declarations are used. This is a prepared submission; official acceptance has not been claimed.